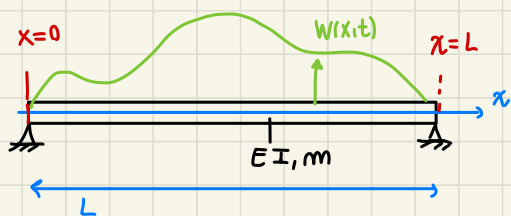


ESE 14 VIBRATION OF DISTRIBUTED PARAMETER SYST

practice about VIBRATIONS / dynamic of distributed parameter syst
on BEAMS, AXIAL / BENDING of SLENDER BEAMS

BENDING VIBRATION OF BEAMS



$$\gamma^4 = \frac{m\omega^2}{EI}$$

SLENDER Beam
bending

single beam with uniform
material / section NO damping and
NO external forces assumptions

↓ STATIONARY
SOLUTION

$$W(x,t) = [A \cos(\gamma x) + B \sin(\gamma x) + C \cosh(\gamma x) + D \sinh(\gamma x)] \cos(\omega t + \varphi)$$

shape of vibration in space

modulation on time

= Ratio between Amplitude / phase $\forall x$ across length

$$W''(x,t) = \gamma^2 [-A \cos(\gamma x) - B \sin(\gamma x) + C \cosh(\gamma x) + D \sinh(\gamma x)] \cos(\omega t + \varphi)$$

(PIN-PIN Beam) what happen when changing the boundaries condition

here we can get an analytical solution of

W (displacement $\perp$ to neutral axis x) $\rightarrow W(x,t)$ STATIONARY

each section moves $\perp$ to x , according to harmonic motion

general solution $W(x,t)$ to specialize respect boundaries condition

we find the solution for PIN-PIN beam (HINGE)

PIN @ $x=0, x=L$ preventing $W(x=0, x=L) = 0$

↳ preventing any transversal
motion

but PIN allows a rotation around it

of the beam, however, by the PIN we get $M=0$ because of dynamic
equilibrium @ $x=0, x=L$ in a section dx on the two ends of the beam

$$M(0,t) = M(L,t) = 0 \quad \forall t$$

and we know the relationship

$$M = \underbrace{EI}_{\text{proportionality coefficient}} \frac{\partial^2 W}{\partial x^2}$$

bending moment
= curvature of the
deformed shape after
bending happens

depend on

material and geometry (I) of section of beam

from $W(x,t)$ expression we are
setting:

$$M=0 \Leftrightarrow \frac{\partial^2 W}{\partial x^2} = 0$$

boundary conditions:

$$(1) \begin{cases} W|_0 = 0 \\ \frac{\partial^2 W}{\partial x^2} \Big|_0 = 0 \end{cases}$$

→ bringing this condition on $W(x,t)$

We end up in a linear system in 4 eq. 4 unknown

A, B, C, D ? 4 unknowns

LINEAR, homogeneous system

↳ NON TRIVIAL SOLUTIONS for this

problem → setting to 0 the det(MATRIX)

$$(2) \begin{cases} W|_L = 0 \\ \frac{\partial^2 W}{\partial x^2} \Big|_L = 0 \end{cases}$$

In this easy case it can be solved

analytically → In any other cases like

CLAMP, FREE end on the Beam, for any mix of

this → the problem is no more solvable

analytically → **numerical solution**

WITH MATLAB

↓
even if analytical
solution exist
we can solve it
numerically anyway

(we can solve also for any other cases through MATLAB)

Writing in full the linear equation from the boundary condition:

valid $\forall t$

$$\begin{cases} [A + C] \cos(\gamma L) = 0 \\ \gamma^2 [-A + C] \cos(\gamma L) = 0 \end{cases} \Rightarrow \begin{cases} +A + C = 0 \\ -A + C = 0 \end{cases}$$

(linear equation in A, C homogeneous)

(leave it generally even if we know that $A = C = 0$)

$$\begin{cases} A \cos(\gamma L) + B \sin(\gamma L) + C \cosh(\gamma L) + D \sinh(\gamma L) = 0 \\ -A \cos(\gamma L) - B \sin(\gamma L) + C \cosh(\gamma L) + D \sinh(\gamma L) = 0 \end{cases}$$

coefficients of the equation, known coefficients (γ known)

linear eq

↑ our linear system of 4 eq in 4 unk →

$$[H(\gamma L)] \underline{x} = 0$$

$$\underline{x} = \begin{bmatrix} A \\ B \\ C \\ D \end{bmatrix} \quad H = \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ \cos(\gamma L) & \sin(\gamma L) & \cosh(\gamma L) & \sinh(\gamma L) \\ -\cos(\gamma L) & -\sin(\gamma L) & \cosh(\gamma L) & \sinh(\gamma L) \end{bmatrix}$$

{ this matrix
can here-worked
to be simplified }

But we [↑] wanna
solve it numerically

once written the problem as Linear homogeneous to find NON SINGULAR solutions $\rightarrow$

$$\det([H(\gamma L)]) = 0$$

But for now γ is unknown...

We can choose γ such that a priori $\det(H) = 0$

$$H = \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ \cos(\gamma L) & \sin(\gamma L) & \cosh(\gamma L) & \sinh(\gamma L) \\ -\cos(\gamma L) & -\sin(\gamma L) & \cosh(\gamma L) & \sinh(\gamma L) \end{bmatrix}$$

looking here rows 3,4 some depends on (γL)

$$H(\gamma L)$$

NON dimensional

$$[\gamma] = [1/m]$$

$$[L] = [m]$$

$\Downarrow$ find solutions of H function

If γ as $\det = 0 \rightarrow$ we can prove this solution can be resolved to

$$\sin(\gamma L) = 0 \Rightarrow \gamma = \frac{k\pi}{L} \quad \begin{matrix} k=1,2,3,\dots \\ k \in \mathbb{N} \end{matrix}$$

analytical solution

you

don't know the easy solution of γ ? $\Rightarrow$ how to find numerically

PROCEDURE:

goes across certain **STEPS:**

STEP 0: define a freq. range for the analysis

(in general the eq is transcendental $\rightarrow \cos \omega L \rightarrow \infty$ nat freq solutions...

if solving numerically we should define a limit of freq! so by the freq that we are interested in) **ω_{MAX}** based on my interest on the problem from γ, ω relation

$\Downarrow$ (if ROBOTIC ARM moving, all $\omega > 300\text{ Hz}$ is not useful!)

$$0 \leq \gamma \leq \gamma_{MAX}$$

(γ_{MAX} related to ω_{MAX})

We should know where to stop

$$\gamma_{MAX} = \sqrt[4]{\frac{m}{EI} \omega_{MAX}^2} = \sqrt{\omega_{MAX} \sqrt{\frac{m}{EI}}}$$

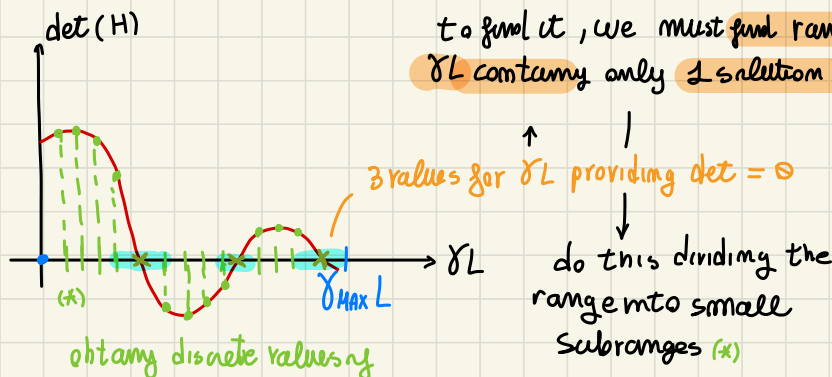
STEP 1: find the special γ values in that range

that are bringing $\det(H) = 0$ (good solution) $\rightarrow$ numerically many ways splitting in 2 STEPS!

isolate the solution natural frequencies

$\hookrightarrow$

Step 1 graphically



→ select the ranges where the two values are of opposite sign (so with neg product)

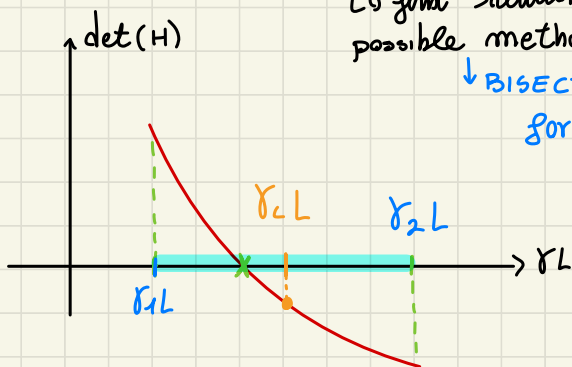
so on that range you have a solution!

← you get as many ranges as the TOTAL amount of solutions of $\det(H(\delta L)) = 0$ inside $(0, \delta_{\max} L)$

↓ so zoom mean the first range

STEP 2

Refine frequencies



to find solution there are multiple possible methods!

↓ BISECTION method for example

on the range $(\delta_1 L, \delta_2 L)$

you take

$\delta_c L$ with

← $\delta_c = \frac{\delta_1 + \delta_2}{2}$ mean (central point)

and you evaluate $\det(H(\delta_c L))$?

• IF: $\det(H(\delta_c L)) \cdot \det(H(\delta_1 L)) < 0$

$\delta_2 \rightarrow \delta_c$ (replace)

still range where $\det(H)$ has opposite sign on the two sides... (divided by 2 the range)

• ELSE:

$\delta_1 \rightarrow \delta_c$

end

(having divided the range in 2 parts) with opposite sign $\det$ on the sides

so we have still the solution inside the new range

repeat this procedure

While $(\gamma_1 L - \gamma_2 L) < \text{tolerance}$

• IF: $\det(H(\gamma_c L)) \cdot \det(H(\gamma_1 L)) < 0$

$\gamma_2 \rightarrow \gamma_c$ (replace)

• ELSE:

$\gamma_1 \rightarrow \gamma_c$

end

end

Repeat this

refine of freq $\forall$ solution
range inside $(0, \gamma_{\max})$

↓

We find the first m in this range
solutions and $\forall \gamma_k$

corresponding
NAT FREQ.

$$\omega_k = \gamma_k^2 \sqrt{\frac{EI}{m}}$$

but $\forall \omega_k$ we

have corresponding modal shape?

How to find the modal shapes...

↓

IF it cannot be solved analytically...

$$[H(\gamma L)] \underline{x} = \underline{0}$$

We plug in one γ_k respect tolerance solution

that correspond to $\det[H(\gamma_k L)] = 0$ so one row is

full of 0 OR a lin comb of the others $\rightarrow$ one row is dependent from the others

IF it is the q^{th} ROW

↓ We remove this row and work
with a reduced $[H]$ matrix of 3 rows and
4 columns (remove dependent row)

reduce the range

until it is so small

we can take the

solution inside that range

↓

{ setting tolerance as a }
{ pure number }

We will find

$$\sin(\gamma L) = 0 \rightarrow \gamma L = k\pi$$

smaller solution is in the order of $\approx \pi$

tolerance $\rightarrow \text{tol} = 1e-4$

we are finding a good
solution enough good

$\text{tol} = 1e(-5)$ as conservativeness
on MATLAB

until now we find
the NATURAL frequencies
 $(0, \omega_{\max})$

↓

3 lim equations left on the system in 4 unknown
 cannot be solved uniquely $\rightarrow$ we should define one of A, B, C, D
 with arbitrary value
 (in fact MODAL shape is define for a scaling factor)

assume $D=1$

$$\underline{\hat{x}} = \begin{bmatrix} A \\ B \\ C \\ 1 \end{bmatrix} \leftarrow \hat{x} \text{ remaining unknown}$$

so we have

$$\begin{bmatrix} \hat{H}(\gamma_{KL}) \end{bmatrix}$$

removing 4TH row (depends on the others)
 and split the first 3 columns from
 the last column

$\underline{N} = 4^{\text{TH}}$ column removed

$$\left[\begin{bmatrix} \hat{H}(\gamma_L) \\ \text{4TH row} \end{bmatrix} - \underline{N} \right] \begin{bmatrix} \hat{x} \\ 1 \end{bmatrix} = \underline{0}$$

$$\Rightarrow \hat{H}(\gamma_L) \hat{x} - \underline{N} = \underline{0}$$

$$\hat{H}(\gamma_L) \hat{x} = \underline{N}$$

FIRST 3 ROW
 3 COLUMNS

is $\det(\hat{H}) \neq 0$

if correctly done, we remove the
 unuseful row (dependent row)

and $D=1$ (for an unknown that can be unitary!)

$$\underline{\hat{x}} = [\hat{H}(\gamma_{KL})]^{-1} \underline{N}$$

so for KRAMER

linear problem!

$$\text{IF } D=2 \quad \underline{\hat{x}} = 2 [\hat{H}]^{-1} \underline{N}$$

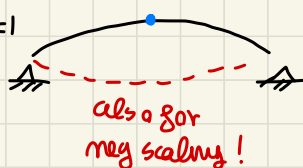
$\underline{x} = \begin{pmatrix} \hat{x} \\ 1 \end{pmatrix}$ is an eigen solution $\forall$ scaling factor of solution

from here we take A, B, C found and $D=1$

plug-in on $W(x,t)$ expression

getting the modal shape of γ_k

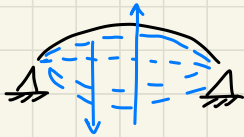
to repeat ! $k=1$



$\forall$ scaling coeff,
 same shape!

(shape) $\times \cos(\omega t + \phi) \rightarrow$ OSCILLATION!

In fact the motion on t is harmonic with time



shape of relative magnitude
as oscillatory motion

simplest in shape on PIN-PIN BEAM

STEP 3 extracting the eigen solutions

assumption... 4th row removed... (it can be wrong) and $D=1$ can be
setted $\rightarrow$ WRONG in our case where $D=0$!

We get on THEPR4 $B \neq 0$ $A=C=D=0$

you can apply any scaling to a NON ZERO value, NOT to a 0 value
that must be 0 always!

Sometimes removing last row and column we can still get
 $\det(\hat{H}) = 0$ NOT SOLVABLE



We should find correct combination of row, column removal
to make $\det(\hat{H}) \neq 0$ of $\hat{H}$ reduced matrix

JUST iterating the procedure checking

$\det(\hat{H})$ norm dim coeff checking which has
Trace(H) $\det$ far 0 and from any choices
possible... I take the one with the
max value (the most far from 0)

and set to 1 that column variable and remove the row



MATLAB coding

SCRIPT "Vib-dps-20221103"

- We set L as GLOBAL variable, so in any function used in the main
script can access the L value of this script $\rightarrow L$ accessible everywhere
- we define A, I also $E, \rho \rightarrow EI, m = \rho A$
 $L = 2m$

defining the name of the function
'simply supported' that evaluate the matrix H from
boundary condition given δL
name func compute H for different δL values

next rows are just checks of definition

then we make the steps

$f_{MAX} = ? \text{INPUT}()$

δx

$\delta L \text{ vector} = [0, \delta_{MAX} L]$ 1000 values inside the range
discretization of STEP 1 of range

- Step 1: isolate-modes perform computation of $\det([H(\delta L)])$
checking conditions of opposite sign $\rightarrow$ find range of solutions
 $i_i = \text{vector of ranges} \equiv \text{array of mat-freq}$
($w_1 \dots w_m$)

• Step 2:

set tolerance $\left\{ \begin{array}{l} \text{NOTICE: all these functions are usable during} \\ \text{the exam! we just need to write} \\ \text{boundary condition} \end{array} \right\}$

iterate over all

solutions in modes setting (δ_1, δ_2) as range of solutions
and refine bisection method

finding (δ_3) a part from tolerance (tol)
solution of $\Rightarrow w_i$

re-evaluate H using function $\text{eval}([--])$

on WORKSPACE

execute command

H is again it self evaluated $\leftarrow$ inside the script

@ new value $\rightarrow$ SINGULAR MATRIX $[H(\delta L)]$

$\downarrow$

then removal of dependent row and find the coefficients

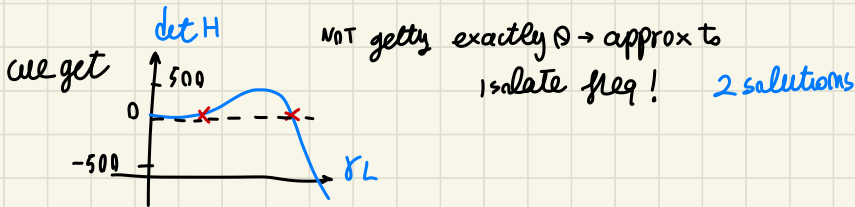
step 3.b: evaluate and draw modal shapes of function

the 'simply supported.m' function JUST get H as
output getting as argument just γL to evaluate $[H(\gamma L)]$
and then definition of $[H]$

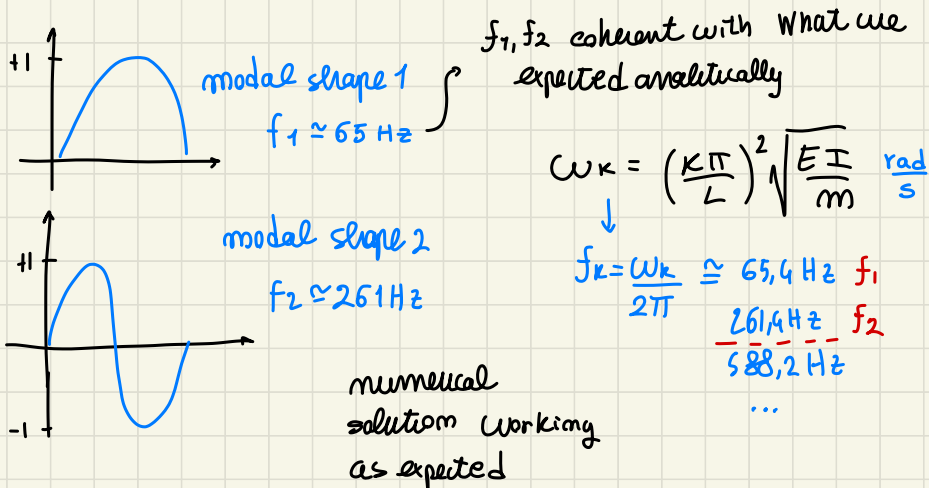
↑ the call of this function by "eval" return
H for new γL value.

to adapt
for new boundaries we just need to redefine H
according to the condition

output: 2 solutions

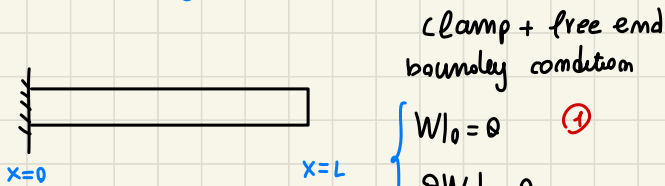


after iteration and isolation we get



ES 2 CANTILEVER BEAM

Slender Beam with uniform material/section properties
(No damping, no forces ecc...)



$$\left\{ \begin{array}{l} W|_0 = 0 \quad (1) \\ \frac{\partial W}{\partial x}|_0 = 0 \quad \text{NO ROTATION} \quad (2) \end{array} \right.$$

implementing this conditions:

$$\left\{ \begin{array}{l} M|_L = 0 \rightarrow \frac{\partial^2 W}{\partial x^2}|_L = 0 \quad (\text{0 bend moment}) \quad (3) \\ T|_L = 0 \rightarrow \frac{\partial^3 W}{\partial x^3}|_L = 0 \quad (\text{No shear force}) \quad (4) \end{array} \right.$$

$$\left\{ \begin{array}{l} (1) \quad A + C = 0 \\ (2) \quad B + D = 0 \\ (3) \quad -A \cos(\gamma L) - B \sin(\gamma L) + C \cosh(\gamma L) + D \sinh(\gamma L) = 0 \\ (4) \quad A \sin(\gamma L) - B \cos(\gamma L) + C \sinh(\gamma L) + D \cosh(\gamma L) = 0 \end{array} \right.$$

↓ new linear homogeneous eq of the new beam boundaries
write the H matrix

$$[H(\gamma L)] = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ -\cos(\gamma L) & -\sin(\gamma L) & \cosh(\gamma L) & \sinh(\gamma L) \\ \sin(\gamma L) & -\cos(\gamma L) & \sinh(\gamma L) & \cosh(\gamma L) \end{bmatrix}$$

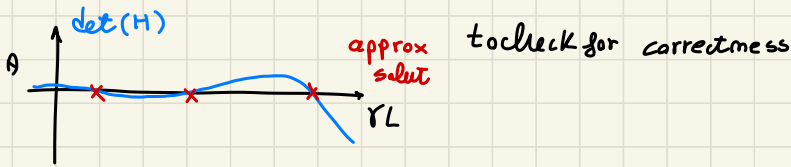
↪ implemented in MATLAB

MATLAB same function structure but 'cantilever' implement
[H] matrix

and on main script JUST use name func = 'cantilever'

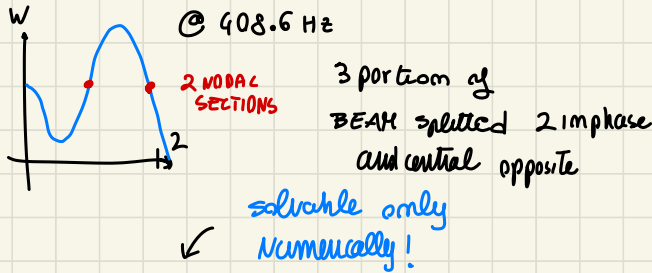
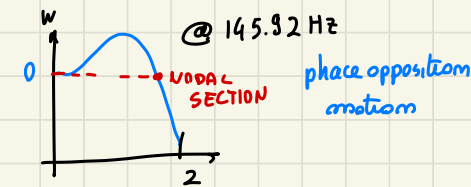
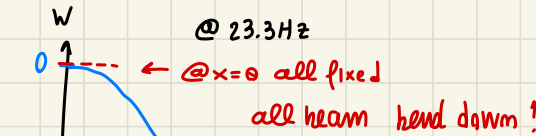
Same script but using the new MATRIX each time

for $f_{MAX} = 6001 \text{ Hz} \rightarrow 3 \text{ mat freq}$



MODAL SHAPE :

same beam!
all change
by a new
boundary
condition!



given by formulae

BOOKS with correction C_K coefficient for $K=1,2,3$

$$W_K = C_K \frac{k^2 \pi^2}{L^2} \sqrt{\frac{EI}{m}}$$

NUMERICAL

comparing with PIN-PIN Beam ratio!

$C_1 \approx 0.356$ Euler beam

$C_2 \approx 0.56$

$C_3 \approx 0.63$ similar

- Just defining t respect boundaries condition (set of boundary)

for others possible combination $\rightarrow$ Use MATLAB to define the solution just by H

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BENDING VIBRATION OF BEAMS

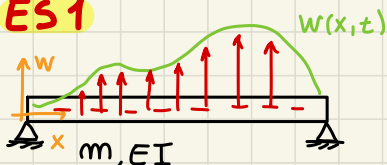
ZUIN INCAPALE

until now we consider continuous bodies... → now deformability on beams

connected to ground by constraints

+ possible springs or Rigid bodies on beam

ES1



we can find analytical solution,
find equation and solve it by MATLAB
to compare numerical-analytical

our objective is find the equations to write matrix of coefficients.

and solution by fixing the zeros of $\det(H)$

↳ But you need to run a code

Matrix of coefficients and other issues

possible
deformation
 $W(x,t)$

giving $\delta x, \delta t$

the normal
beam displacement

↓ with expression:

for STEEL

$$E = 2.1 \cdot 10^{11} \frac{N}{m^2}$$

ALUMINIUM

$$E = 7 \cdot 10^{10} \frac{N}{m^2}$$

$$m = M/L \quad [m] = [kg/m]$$

EI : bending stiffness known

Young's modulus
moment of inertia
respect neutral axis

$$I = \frac{1}{12} b \cdot s^3$$

$$I = \frac{\pi}{64} D^4$$

$$[I] = [m^4] \quad \text{measure unit}$$

$$W(x,t) = [A \cos(\delta x) + B \sin(\delta x) + C \cosh(\gamma x) + D \sinh(\gamma x)] \cos(\omega t + \varphi)$$

↑ unknown → we need 4 equations
(2 $\forall$ p.m.)

one function
describing ALL

bending vibration... In case of DISCONTINUITY on the beam we have to
add more unknowns, have more functions

$$\frac{\partial W(x,t)}{\partial x} = W'(x,t) = \gamma [-A \sin(\delta x) + B \cos(\delta x) + C \sinh(\gamma x) + D \cosh(\gamma x)] \cos(\omega t + \varphi)$$

$$W''(x,t) = \gamma^2 [-A \cos(\delta x) - B \sin(\delta x) + C \cosh(\gamma x) + D \sinh(\gamma x)] \cos(\omega t + \varphi)$$

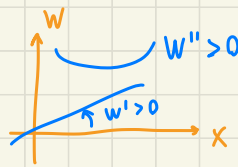
sign convention

$W > 0$ of upwards



W' is the ROTATION of the displacement, $W' > 0$ if
counterclockwise, to check that initial
rotation is 0, we use $W'(x,t)$ @ x inserted
here with HINGE μ bending moment = 0

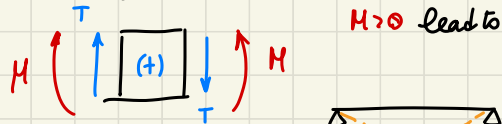
and M depends on $\frac{\partial^2 W}{\partial x^2}$



positive convention of W', W'' sign
on bending moment
 M respect W'' depends on sign convention

$$M(x,t) = EI \frac{\partial^2 W}{\partial x^2} = EI W'' \quad \leftarrow$$

respect sign convention

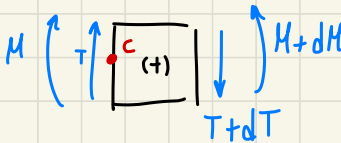


$H > 0$ lead to

positive curvature of bendy $\Rightarrow$ this deformed shape of also, tension on below compress on up

$$T(x,t) = \frac{\partial M(x,t)}{\partial x} = EI \frac{\partial^3 W}{\partial x^3} = EI W'''$$

(easy to find formula of infinitesimal section)



T, M rises along x

equilibrium of moments respect C

$$M - M - dM + (T + dT)dx = 0$$

$$-dM + Tdx + \cancel{dTdx} = 0$$

negligible

with this sign convention

$$\rightarrow T = dM/dx \text{ as expected}$$

IMPORTANT..

when add additional elements like springs or RIGID BODIES

expressing all derivative of W, W', W'', W''' useful

$$W'''(x,t) = \gamma^3 [A \sin(\gamma x) - B \cos(\gamma x) + C \sinh(\gamma x) + D \cosh(\gamma x)] \cos(\omega t + \varphi)$$

need for shear force

and the relationship between

$$\gamma^4 = \frac{\omega^2 m}{EI}$$

Now we need conditions for the equation to write $A, B, C, D, \delta \rightarrow W$
 check the ENDS of the BEAM

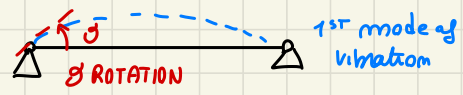
2 conditions $\forall$ side ... HINGE

same boundary:

$W(x, t) @ x=0$ of HINGE! $\forall t$

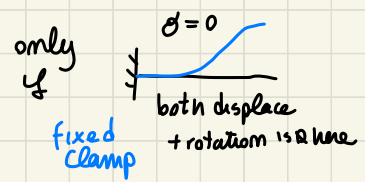
1) $W(0, t) = 0 \quad \forall t \quad (A + C) \cos(\omega t + \varphi) = 0$
 (to be valid $\forall t$)

2) the rotation @ A is $\neq 0$ because
 there is an HINGE, $W' \neq 0$ IT CAN ROTATE



7IN allows rotation! (unknown)
 (normalized mode of vibration)

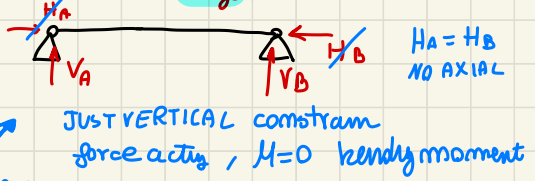
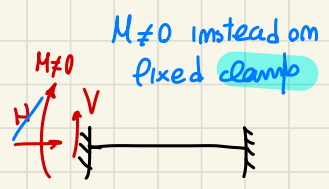
No condition on $W' @ x=0$



checking for $M \cong W''$ it is known? so we can write an equation

because there is an hinge $M=0 \rightarrow$

an Hinge has JUST this constraints



WE WORK on this condition

$M(0, t) = 0 = \cancel{\gamma}^2 [-A - C] \cos(\omega t + \varphi) = 0$
 $\forall t$